Interface Design

DESIGN

TASK CHECKLIST

- [x] Select Design Strategy.
- [x] Design Architecture.
- [x] Select Hardware and Software.
- [ ] Develop Use Scenarios.
- [ ] Design Interface Structure.
- [ ] Develop Interface Standards.
- [ ] Design Interface Prototype.
- [ ] Evaluate User Interface.
- [ ] Design User Interface.
- [ ] Develop Physical Data Flow Diagrams.
- [ ] Develop Program Structure Charts.
- [ ] Develop Program Specifications.
- [ ] Select Data Storage Format.
- [ ] Develop Physical Entity Relationship Diagram.
- [ ] Denormalize Entity Relationship Diagram.
- [ ] Performance Tune Data Storage.
- [ ] Size Data Storage.

User interface (UI) is all the hardware, software, screens, menus, functions, outputs, and features that enable **two-way communications** between the user and the computer.

The user interface is the key to **usability**.

What is Usability?

The official ISO 9241-11 definition:
“the extent to which a product can be used by specified users to achieve specified goals with **effectiveness**, **efficiency** and **satisfaction** in a specified context of use.”

i.e. : A usable system is a system that is **easy to use** and **easy to learn**.

Effectiveness

- ❑ Does it do the right thing?
- ❑ Does it get things done?

Efficiency

- ❑ Does it do that with minimum (physical and mental) effort?

Satisfaction

- ❑ Does it make you feel good?
- ❑ Do you feel happy?

A usable interface

1. **Easy** to **become familiar with and competent in using during the first contact**

   → e.g if a travel agent's website is a well-designed one, the user should be able to move through the sequence of actions to book a ticket **quickly**.

2. **Easy to achieve objectives**

   → If a user has the goal of booking a flight, a good design will guide him/her through the **easiest process** to purchase that ticket.

3. **Easy to recall** the user interface and how to use it on **subsequent visits**.

   → the user should learn from the first time and book a second ticket just as easily.

A Transparent Interface

- The best interfaces are the ones that users do not even notice
- They do what the user expects them to do
- They are effectively **invisible**
- They are **intuitive**
- They just work!
- They **do not distract** the users and and call attention to itself.
- They create the illusion that users aren’t interacting with a device so much as they’re trying to attain goals **directly** and as effortlessly as possible.

The Process (Of Designing User Interface)

Interface Design Process

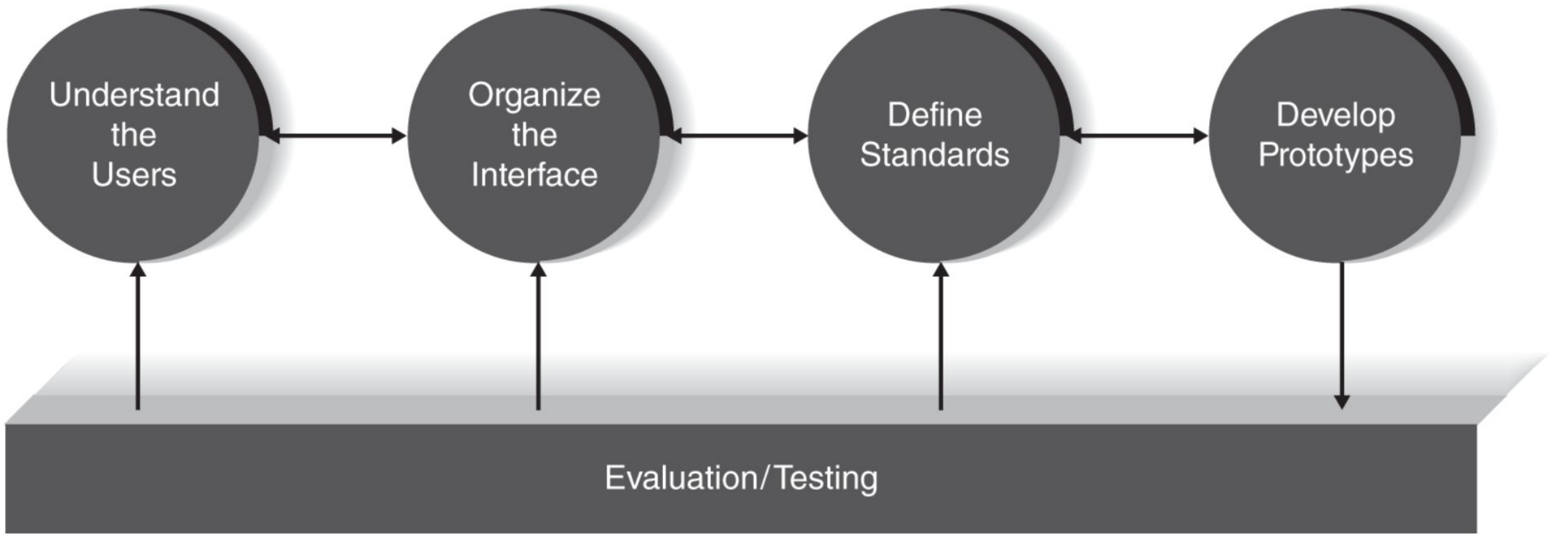

FIGURE 8-7 Interface design process.

Understand the Users

- **Human Centered Design** argues that when we use words that make people sound like another component of the system (words like user), we are in a sense de-humanizing them.
- Since we need to put a label on users, we use **personas** when developing designs.
- Personas are fictional characters that reflect user types, pinpointing who they are and what they do in relevant **contexts**
- Persona's represent **groups of users** not a single user.

DronTeq’s Personas

Martin: wants to learn details about DrōnTeq's products	
“I’m focused on finding the best commercial drone for my business needs.”	
Knowledge:	Understands commercial drones and has specific uses and capabilities in mind for the drone purchase.
Tasks:	Arrives a site to select and configure custom drones. Wants to be able to easily identify drone models and options to meet his intended purposes. Wants to save and compare several configurations in terms of capabilities and prices.
Interests:	Has a business interest in commercial drones. Wants to find the best tool to use to fulfill business needs. Cost/benefit analysis is important.
Characteristics:	Deliberate; thoughtful; expects to be able to develop and compare alternative configurations easily and seamlessly.
Jesse: wants to learn about the uses of commercial drones	
“I’m intrigued by the possibilities of commercial drones and want to learn more.”	
Knowledge:	Extremely limited specific knowledge of commercial drone uses and capabilities. Most likely knowledgeable about hobby-level drones. Wants to explore through examples, case studies, video clips, and owner descriptions of commercial drone applications.
Tasks:	Arrives at site and looks for video clips of drones performing commercial operations: photography, agricultural surveys, pipeline inspections, security surveillance, etc. Within each usage category, wants to discover more details through product descriptions, examples, and owner comments.
Interests:	Has a general interest in commercial drone applications and prefers to see an array of ideas and examples.
Characteristics:	Curious, open-minded; creative, exploratory. Wants to get excited by the possibilities offered by commercial-level drones.

FIGURE 8-8 DrōnTeq Sales System personas.

The **key feature** of user centered design is **direct engagement** with **real users** to ensure suitability of the solution in **real context of use** not the ones we are imagining.

Context of Use (Use Scenario)

- Common uses of the system.
  - E.g. Focused Visitor vs Exploratory Visitor of DronTeq

Use Scenario: Focused Visitor	Use Scenario: Exploratory Visitor
User wants to easily develop alternative drone configurations and compare price and performance of the alternatives	User wants to learn about applications of commercial drones through video clips and written material
1. User selects a drone base model 2. User looks at lists of available options for that model 3. User selects options and wants to learn price and capabilities of that configuration 4. User saves, modifies, or discards that configuration and repeats with other options or other drones 5. User has one or more configurations to save and consider for purchase. 6. User may want to contact a sales representative with questions.	1. User views video clips of drones being used in commercial applications 2. User selects an application area and views more clips in that area 3. User reads site marketing material 4. User reads owner descriptions and comments 5. User repeats steps 2-4 for other commercial application areas. 6. User views list of drone models 7. User selects a model and reads site marketing material for that drone.

FIGURE 8-9 Two use scenarios for the DrōnTeq sales website.

The **context of use(or use scenarios)** are everything about how systems are **actually employed** in the real world.

Context of Use - Example

- Point-of-Service (POS) system at two different contexts: a restaurant/cafe vs. a grocery store
  - The equipment and software are very similar at the two locations.
  - But the actual use of the systems are very different
  - Supermarket:
    - One User → the checkout person
    - One customer at a time / a lot of items / needs to be done pretty efficiently
    - Typically no need to change to a new customer mid-way through the process
  - Restaurant /Cafe
    - Many Users → servers / bartenders
    - Many customers
    - The system needs to switch between different customers and user very quickly

Basic Principals

- You need to **research**, understand and interact with real users
- **Observation** is the most effective
- questionars, interviews can also be used for research
- Produce **prototypes** as early as possible
    - Low fidelity prototypes, storyboards and even paper prototypes can be used
- We need timescales to **allow feedback and design changes**

Interface Design Prototyping

- Defining the **Look**
    - Wire**frame** Diagram
- Defining the **Flow**
    - Storyboard
    - Wire**flow** Diagram
    - HTML prototype
    - Language prototype

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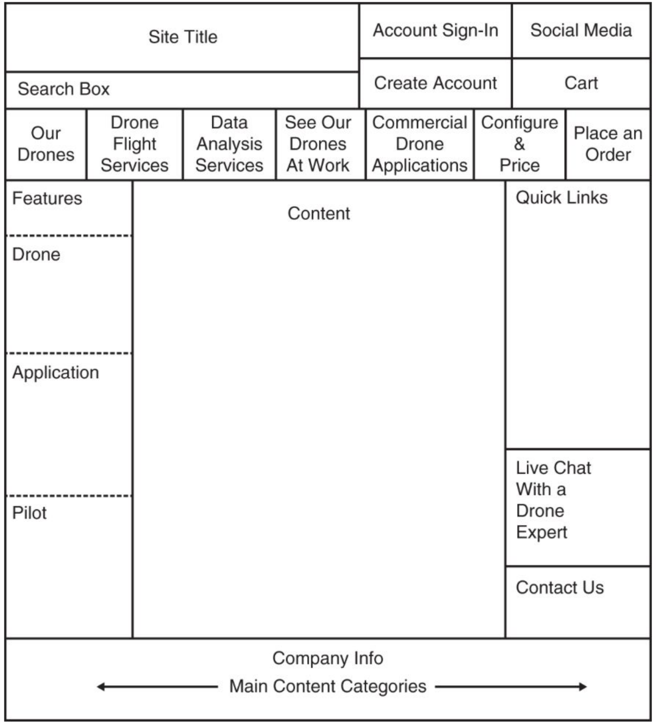

FIGURE 8-12 DrōnTeq wireframe diagram.

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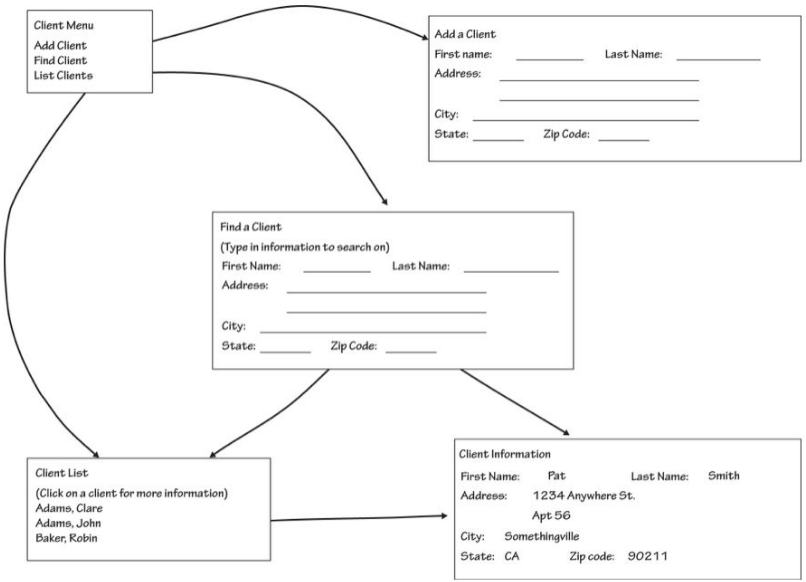


FIGURE 8-13 An example storyboard.

Interface Evaluation/Testing

- Heuristic evaluation
  - Compare to a set of heuristics/principles
- Walk-through evaluation
  - The project team presents the prototype
- Interactive evaluation
  - The users themselves work with HTML or language prototypes
- Formal usability testing
  - Used only with language prototypes. In a special lab equipped with cameras and special software to record keystrokes and mouse operations.

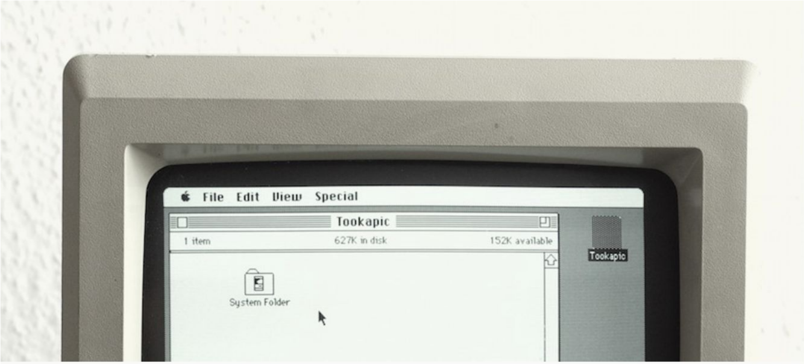
Shneiderman’s Eight Golden Rules

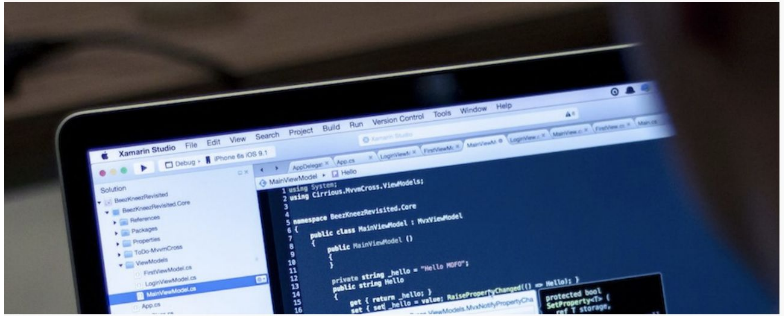
8 Golden Rules of Interface Design

1. Strive for **consistency**.
2. Seek **universal usability**.
3. Offer informative **feedback**.
4. Design dialogs to yield **closure**.
5. Prevent **errors**.
6. Permit easy **reversal** of actions.
7. Keep users in **control**.
8. Reduce short-term **memory load**.

Consistency

- Consistent **sequence of actions** in similar situations
- Identical **terminology** in prompts, menus, and help screens
- Consistent **color, layout, capitalization, fonts** throughout
- **Exceptions** should be comprehensible and limited in number





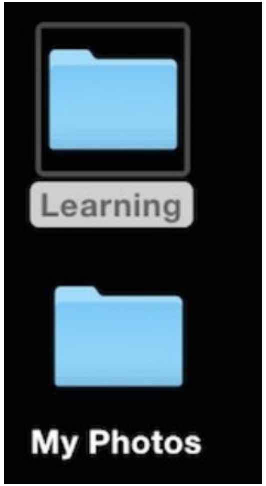
The look of Mac OS over time. Mac OS Menu Bar stays consistent.

Universal Usability

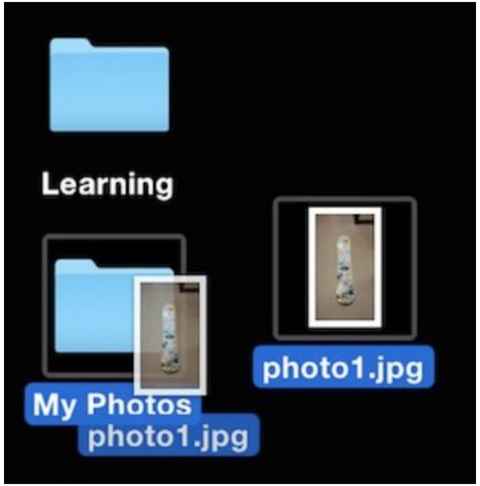
- Recognize the needs of **diverse users**
  - E.g. novice to expert differences, age ranges, disabilities, international variations, technological diversity
- Design for **plasticity and the transformation of content**
- Adding features for novices, e.g. explanations, and features for experts, e.g. shortcut and faster pacing, improves the interface design and **perceived quality**

Informative Feedback

- **For every user action**, there should be an interface feedback
    - Frequent & minor actions → modest feedback
    - Infrequent & major actions → more substantial feedback
- Visual presentation of the objects of interest provides a convenient environment for showing changes explicitly (**Visual Feedback**)



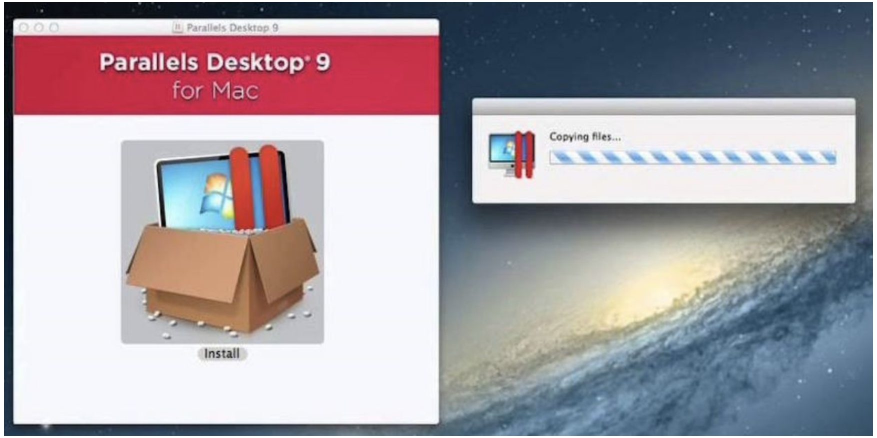
Modest feedback



More substantial feedback

Closure

- Sequences actions should be organized into groups with a **beginning, middle and end**.
- **Informative feedback at the completion** of a group of actions, gives users the **satisfaction** of **accomplishment**, a sense of **relief**, a signal to drop contingency plans and to prepare to the next group of actions.

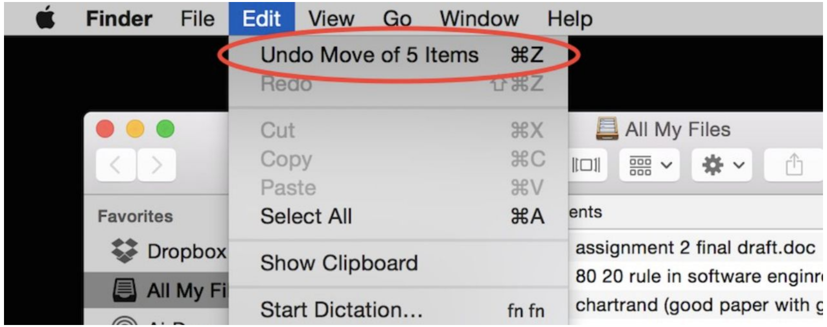


Prevent Errors

- Design so that users **cannot make serious errors**. e.g.
    - Grey out menu items that are no appropriate
    - Do not allow alphabetical characters in numeric fields
- Offer simple, constructive, and specific **instructions** for recovery in case user makes an error
    - Erroneous actions should **leave the interface state unchanged**, or the interface should give instructions about restoring the state
    - **Example:** users should not have to retype an entire name-address form if they enter an invalid zip code but rather should be guided to repair only the faulty part

Reversal of Actions

- As much as possible, **actions should be reversible**
  - Relieves **anxiety**
  - Encourages **exploration**
- **Unit of reversibility**: may be a single action, a data-entry task, or a complete group of actions



Keep Users in Control

- Experienced users **strongly desire** the sense that they are **in charge** of the interface and that the interface responds to their actions.
- They don’t want **surprises** or **changes** in familiar behaviours
- They are **annoyed** by
  - **tedious** data-entry sequences
  - Difficulty obtaining necessary **information**
  - Inability to produce their desired **result**

Reduce Short-Term Memory Load

- **Rule of thumb**: People can remember seven plus or minus two chunks of information.
- Avoid interfaces in which users must remember information from one display and then use that information on another display.
  - E.g. cell phones should not require re-entry of phone numbers
  - Lengthy forms should be compacted to fit a single display